

# ECO 310: Precept 2

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$$C_2 = (1 + r)(100 - C_1).$$

$\implies$  Spending  $C_1$  today means saving  $100 - C_1$ , which grows to  $(1 + r)(100 - C_1)$  for tomorrow's consumption.

**What's the consumer's optimal choice?**

# Intuitive Method (Step 1: Rearranging the Budget Constraint)

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Rearranged form:

$$(1 + r)C_1 + C_2 = 100(1 + r).$$

This representation allows us to identify the implicit prices for  $C_1$  and  $C_2$ .

## Intuitive Method (Step 2: Prices & Marginal Utilities)

- From the rearranged constraint, the coefficients on the consumption goods are their implicit prices:

$$p_1 = 1 + r \quad \text{and} \quad p_2 = 1.$$

- We saw in class with the pens-pencils example that corner solutions are usually optimal with perfect substitutes.
- The utility function is

$$u(C_1, C_2) = C_1 + 0.5C_2,$$

so the marginal utilities are:

$$MU_1 = 1 \quad \text{and} \quad MU_2 = 0.5.$$

- Therefore, the marginal rate of substitution (MRS) is:

$$\frac{MU_1}{MU_2} = \frac{1}{0.5} = 2.$$



## Intuitive Method (Step 3: Comparing Ratios)

Compare the MRS with the price ratio:

$$\frac{p_1}{p_2} = 1 + r.$$

- **If  $2 > 1 + r$  (i.e.,  $r < 1$ ):** The consumer gets more utility per dollar from consuming today. Thus, a corner solution is optimal with  $C_2 = 0$  and all spending on  $C_1$ .
- **If  $2 < 1 + r$  (i.e.,  $r > 1$ ):** The consumer prefers tomorrow's consumption, so  $C_1 = 0$  and all spending goes to  $C_2$ .
- **If  $r = 1$ :** The consumer is indifferent, and any bundle on the budget line

$$(1 + r)C_1 + C_2 = 100(1 + r)$$

is optimal.

# Intuitive Method: Summary of Results

The optimal consumption bundles are:

$$(C_1, C_2) = \begin{cases} (100, 0) & \text{if } r < 1, \\ (0, 100(1 + r)) & \text{if } r > 1. \end{cases}$$

And if  $r = 1$ , any bundle satisfying

$$(1 + r)C_1 + C_2 = 100(1 + r)$$

is optimal.

# Lagrangian Method (Step 1: Setting Up)

We now solve the problem using the Lagrangian method (for the case  $r < 1$ ).

**Objective:** Maximize

$$u(C_1, C_2) = C_1 + 0.5C_2.$$

**Constraints:**

- $C_1 \geq 0$
- $C_2 \geq 0$
- Budget constraint:  $C_2 \leq (1 + r)(100 - C_1)$ .

**Lagrangian:**

$$L = C_1 + 0.5C_2 + \lambda_1 C_1 + \lambda_2 C_2 + \lambda_3 (100(1 + r) - (1 + r)C_1 - C_2).$$

## Lagrangian Method (Step 2: Identifying Binding Constraints)

Based on our intuitive analysis (for  $r < 1$ ):

- The budget constraint binds (i.e., it holds with equality).
- We expect  $C_2 = 0$  (corner solution), so the non-negativity constraint on  $C_2$  is **binding**. Without this rule, the consumer would make  $C_2$  negative and use this to finance a large  $C_1$ . So consumer will respond to this unfortunate rule by satisfying constraint with equality,  $C_2 = 0$ , and multiplier  $\lambda_2$  is positive, meaning a real restriction.
- The constraint  $C_1 \geq 0$  is **non-binding** because the consumer will choose  $C_1 > 0$ . We think the consumer wants a positive quantity anyway, so the rule that they can't go negative has no impact. So set  $\lambda_1 = 0$  (ignore the constraint) and verify that it's satisfied anyway.

# Lagrangian Method (Step 3: FOCs & Complementary Slackness)

Assuming our guess is correct, we write the first-order conditions (FOCs):

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$$\frac{\partial L}{\partial C_2} = 0.5 + \lambda_2 - \lambda_3 = 0.$$

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## Complementary Slackness:

- For  $C_1 \geq 0$ :  $\lambda_1 = 0$  (non-binding).
- For  $C_2 \geq 0$ : Since  $C_2 = 0$  (binding), we must have  $\lambda_2 > 0$ .
- The budget constraint binds, so  $\lambda_3 > 0$ .

# Lagrangians (FOCs and Complementary Slackness)

- Now, write out the 5 equations (FOC's and complementary slackness conditions), assuming our guess is right:

FOC for  $C_1$ ,  $\frac{\partial L}{\partial C_1} = 0$  : (1)  $1 + \lambda_1 - (1 + r)\lambda_3 = 0$

FOC for  $C_2$ ,  $\frac{\partial L}{\partial C_2} = 0$  : (2)  $\frac{1}{2} + \lambda_2 - \lambda_3 = 0$

non-binding  $C_1 \geq 0$  constraint: (3)  $C_1 \geq 0$  and  $\lambda_1 = 0$

binding  $C_2 \geq 0$  constraint: (4)  $C_2 = 0$  and  $\lambda_2 > 0$

binding budget constraint: (5)  $C_2 = (1 + r)(100 - C_1)$  and  $\lambda_3 > 0$

# Lagrangian Method (Step 4: Solving the Equations)

Solve the equations step by step:

- ① From the FOC for  $C_1$ :

$$1 - (1 + r)\lambda_3 = 0 \implies \lambda_3 = \frac{1}{1 + r}.$$

- ② From the FOC for  $C_2$ :

$$0.5 + \lambda_2 - \lambda_3 = 0 \implies \lambda_2 = \lambda_3 - 0.5 = \frac{1}{1 + r} - 0.5.$$

- ③ Since  $C_2 = 0$  (from the binding non-negativity constraint) and using the budget constraint:

$$C_2 = (1 + r)(100 - C_1) \implies 0 = (1 + r)(100 - C_1),$$

it follows that:

$$C_1 = 100.$$

## Lagrangian Method (Step 5: Final Check & Conclusion)

Our solution is:

$$C_1 = 100, \quad C_2 = 0, \quad \lambda_1 = 0, \quad \lambda_2 = \frac{1}{1+r} - 0.5, \quad \lambda_3 = \frac{1}{1+r}.$$

For the multipliers to be valid (i.e., non-negative), we require:

$$\frac{1}{1+r} - 0.5 > 0 \iff \frac{1}{1+r} > 0.5 \iff r < 1.$$

This condition is consistent with our assumption. Hence, the solution is confirmed.

## Question 4: Marginal Rate of Substitution

- 1 Consider the utility function  $u(C, T) = (4C + T)^2$ . Explain why this person is always willing to trade one coffee for four teas.
- 2 Explain why if  $p_C < 4p_T$ , a utility-maximizing consumer with this utility will spend all her money on coffee.

# Answer to Question 4 (1)

## Trade-off Rate (MRS):

- Recall:

$$MRS_{\text{coffee for tea}} = - \left. \frac{\Delta \text{coffee}}{\Delta \text{tea}} \right|_{\text{indiff}} = \frac{MU_{\text{tea}}}{MU_{\text{coffee}}}.$$

- Note that  $u(C, T)$  is a monotonic transformation of  $v(C, T) = 4C + T$ . They represent the same preferences.
- From  $v(C, T) = 4C + T$ :

$$MU_{\text{coffee}} = 4, \quad MU_{\text{tea}} = 1.$$

Hence

$$\frac{MU_{\text{tea}}}{MU_{\text{coffee}}} = \frac{1}{4},$$

i.e. one coffee is worth exactly four teas to this consumer.

## Answer to Question 4 (2)

### All Spending on Coffee if $p_C < 4p_T$ :

- The consumer's marginal rate of substitution is constantly 4 (coffee is always 4 times as valuable as tea).
- If coffee costs less than 4 times what tea costs ( $p_C < 4p_T$ ), then the “bang per buck” for coffee exceeds that for tea:

$$\frac{MU_C}{p_C} > \frac{MU_T}{p_T}.$$

- Thus a utility-maximizing consumer should allocate *all* spending to coffee.